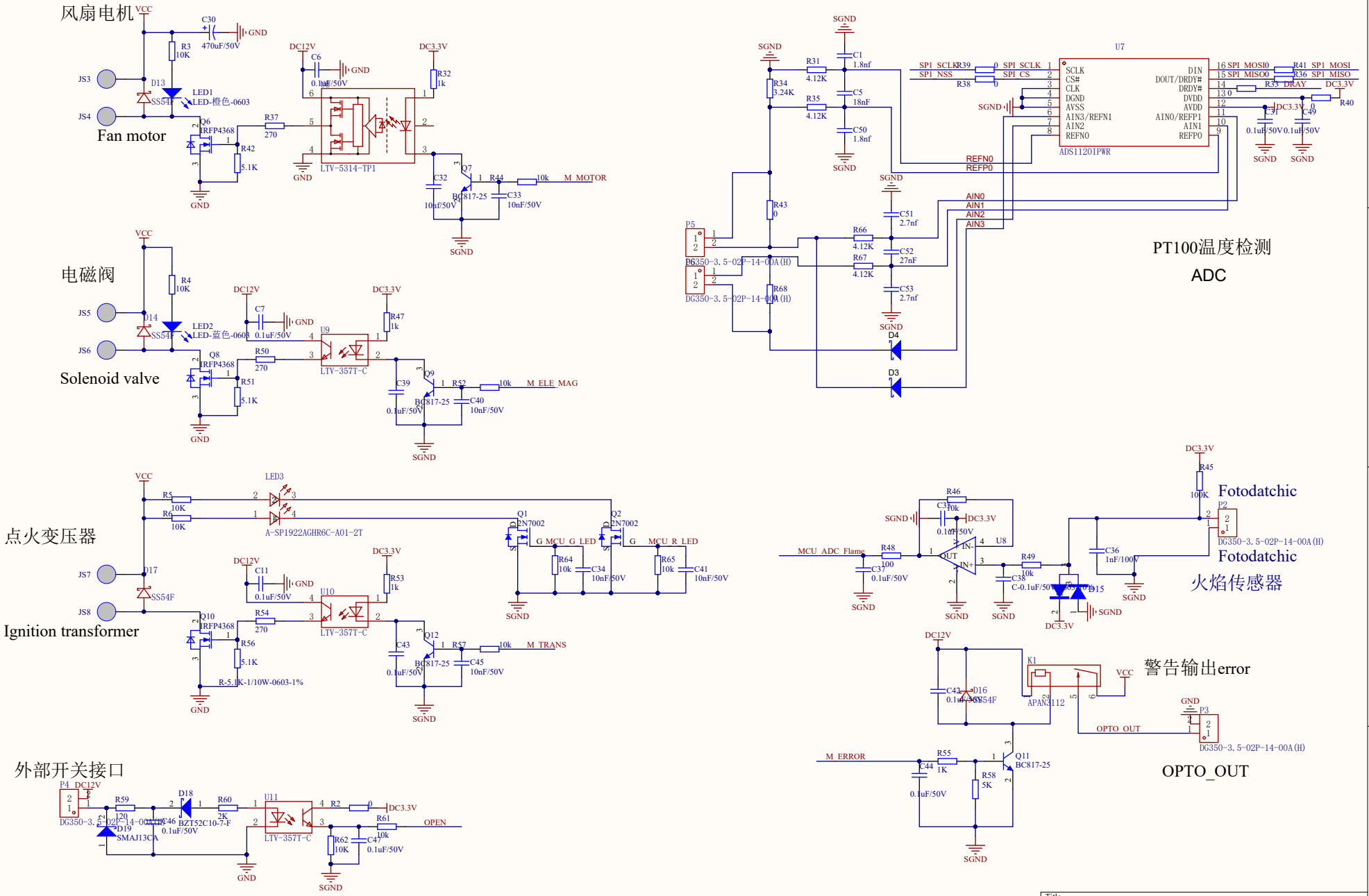
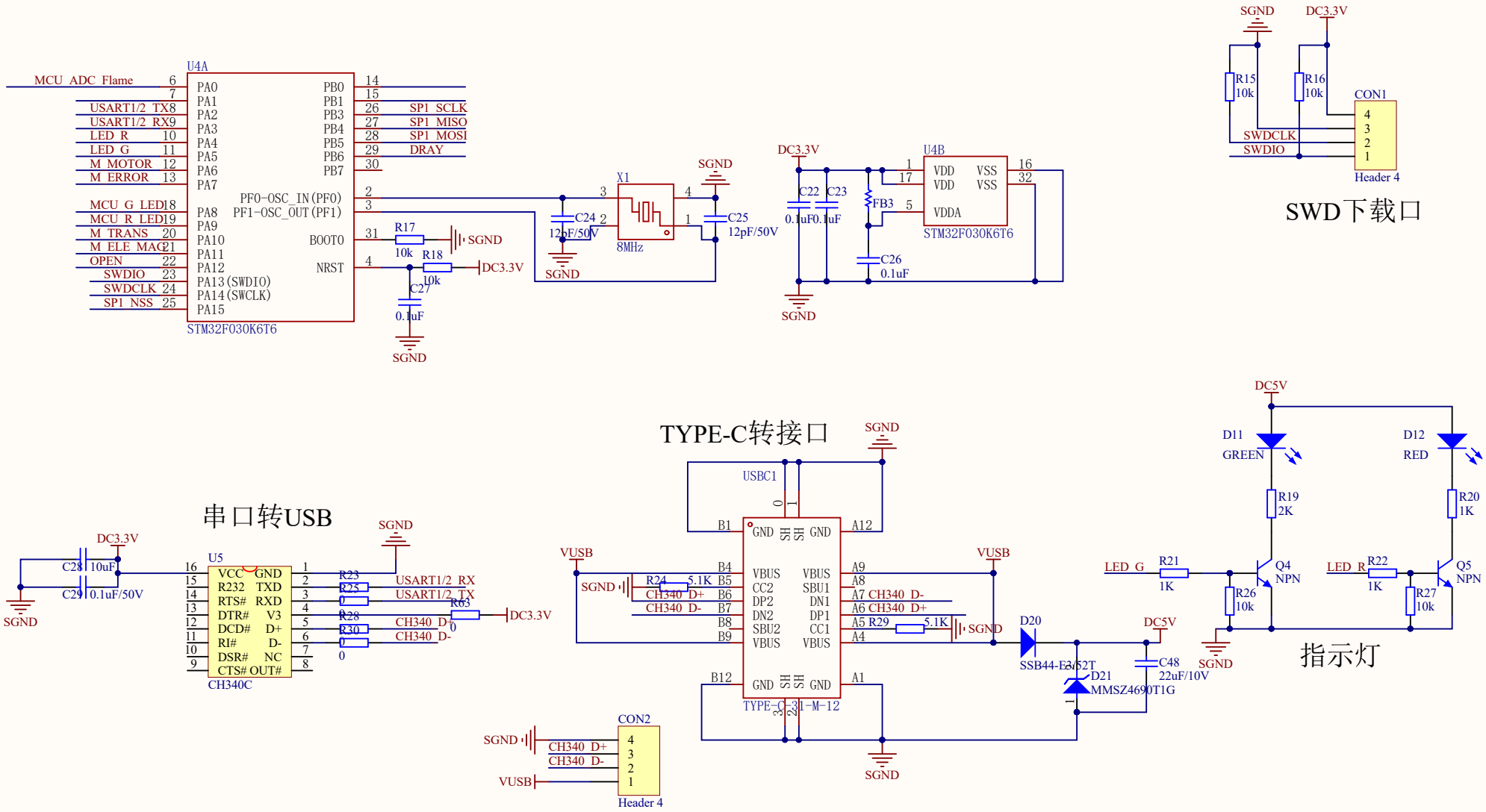
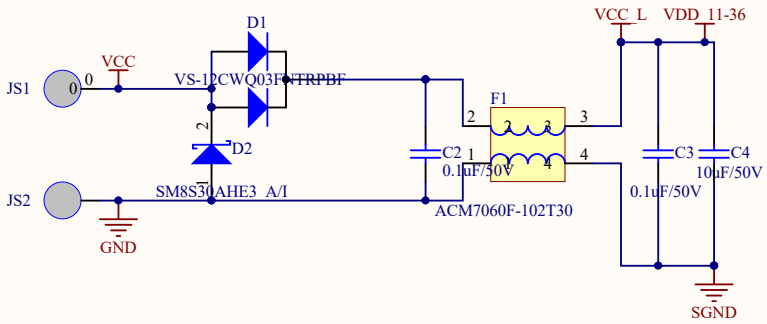
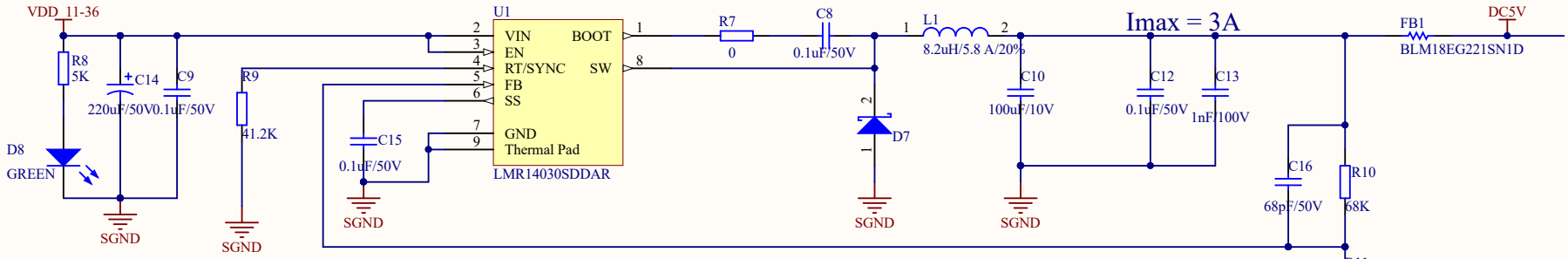

风扇电机
Fan motor
电磁阀
Solenoid valve
点火变压器
Ignition transformer
外部开关接口
PT100温度检测
ADC
Fotodatchic
Fotodatchic
火焰传感器
警告输出error
OPTO_OUT
Title
Size
A3
Number
Revision
Date: 2025/7/30
Sheet of
File: G:\6、其他文件\..\电磁控制-interface-PCB.SchDoc
Drawn By:


U4A
MCU_ADC_Flame
USART1/2_TX
USART1/2_RX
LED_R
LED_G
M_MOTOR
M_ERROR
MCU_G_LED
MCU_R_LED
M_TRANS
M_ELE_MAG
OPEN
SWDIO
SWDCLK
SP1_NSS
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(SWDIO)
PA14(SWCLK)
PA15
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PF0-OSC_IN(PF0)
PF1-OSC_OUT(PF1)
BOOT0
NRST
SP1_SCLK
SP1_MISO
SP1_MOSI
DRAY
STM32F030K6T6
X1
8MHz
C24
12pF/50V
C25
12pF/50V
R17
10k
R18
10k
C27
0.1uF
SGND
DC3.3V
U4B
VDD
VSS
VDDA
C22
C23
0.1uF
FB3
C26
0.1uF
STM32F030K6T6
SWD下载口
CON1
Header 4
R15
10k
R16
10k
SWDCLK
SWDIO
串口转USB
U5
VCC
GND
R232
TXD
RTS#
RXD
DTR#
V3
DCD#
D+
RI#
D-
DSR#
NC
CTS#
OUT#
CH340C
C28
10uF
C29
0.1uF/50V
R23
R25
R28
R30
R63
USART1/2_RX
USART1/2_TX
CH340_D+
CH340_D-
TYPE-C转接口
USBC1
TYPE-C-31-M-12
VUSB
R24
5.1K
CH340_D+
CH340_D-
R29
5.1K
D20
SSB44-E3/52T
D21
MMSZ4690T1G
C48
22uF/10V
CON2
Header 4
D11
GREEN
D12
RED
R19
2K
R20
1K
R21
1K
R22
1K
R26
10k
R27
10k
Q4
NPN
Q5
NPN
LED_G
LED_R
指示灯
Title
Size
A4
Number
Revision
Date:
2025/7/30
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File:
G:\6、其他文件\..\电磁控制-MCU-P2.SchDoc
By:



输入电压范围在11-39V之间，超过这个范围电路保护不供电



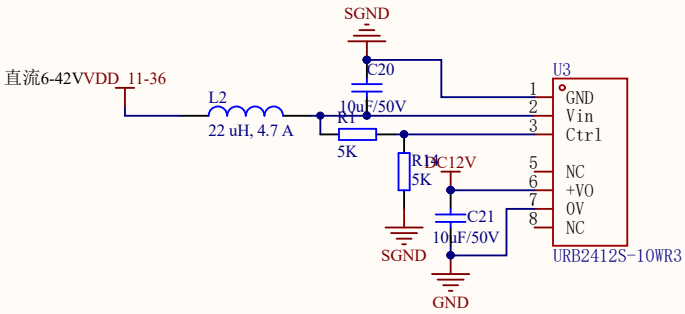
Vout=Vref * (1 + Rfbt/Rfbb)
=Vref * (1 + R19/R20)
=0.75V* (1 + 68K/12.1K)
=4.965V

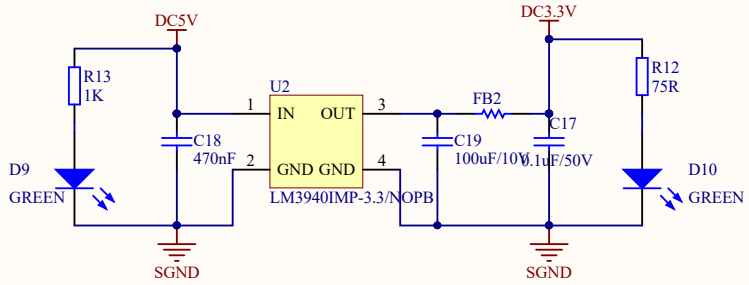
Switch frequency is about 600KHz

$R_T(kOhm) = 32537 * f_{SW}(kHz)^{-1.045}$

$L > [(Vin-Vout)/Iout*K_{IND}]*Vout/(Vin*f_{SW}) = 5.497uH$

$K_{IND} = 40\%$





Title		
Size A4	Number	Revision
Date: 2025/7/30		Sheet of
File: G:\6、其他文件\..\电磁控制-POWER修改.PrjPcb		Drawn By: